4. Show that the following points taken in order form an equilateral triangle in each case.
(i) $A(2, 2), B(-2, -2), C(-2\sqrt{3}, 2\sqrt{3})$ (ii) $A(\sqrt{3}, 2)$, B (0,1), C(0,3)

5. Show that the following points taken in order form the vertices of a parallelogram.
(i) $A(-3, 1), B(-6, -7), C(3, -9), D(6, -1)$ (ii) $A(-7, -3)$, B(5,10), C(15,8), D(3, –5)

6. Verify that the following points taken in order form the vertices of a rhombus.
(i) $A(3,-2), B(7,6), C(-1,2), D(-5, -6)$ (ii) A (1,1), B(2,1),C (2,2), D(1,2)

7. A (–1, 1), B (1, 3) and C (3, a) are points and if $AB = BC$, then find 'a'.

8. The abscissa of a point A is equal to its ordinate, and its distance from the point $B(1, 3)$ is 10 units, What are the coordinates of A?

9. The point (x, y) is equidistant from the points (3,4)and(–5,6). Find a relation between x and y.

10. Let $A(2, 3)$ and $B(2,-4)$ be two points. If P lies on the x-axis, such that $AP = \frac{3}{7} AB$, find the coordinates of P.

11. Show that the point (11,2) is the centre of the circle passing through the points (1,2), (3,–4) and (5,–6)

12. The radius of a circle with centre at origin is 30 units. Write the coordinates of the points where the circle intersects the axes. Find the distance between any two such points.

5.4 The Mid-point of a Line Segment


Fig. 5.20

Imagine a person riding his two-wheeler on a straight road towards East from his college to village A and then to village B. At some point in between A and B, he suddenly realises that there is not enough petrol for the journey. On the way there is no petrol bunk in between these two places. Should he travel back to A or just try his luck moving towards B? Which would be the shorter distance? There is a dilemma. He has to know whether he crossed the half way mid-point or not.



Fig. 5.21

The above Fig. 5.21 illustrates the situation. Imagine college as origin O from which the distances of village A and village B are respectively x_1 and x_2 $(x_1 < x_2)$. Let M be the mid-point of AB then x can be obtained as follows.

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